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JNANA SANGAMA, BELAGAVI - 590014

A Project Report on

"MULTI-LANGUAGE INTERACTIVE TICKET DISPENSING ROBOT"

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING IN

ROBOTICS AND AUTOMATION

ENGINEERING

Submitted by:

Ms. VAISHNAVI V PATIL	**2AG23RA413**
Ms. BHAVANI V REVANKAR	**2AG23RA401**
Ms. SHILPA S SHETTAR	**2AG22RA008**
Ms. SOUMYA S BIRADAR	**2AG22RA010**

Under the Guidance of

Dr. Pooja Angolkar

Professor and Head, Department of Robotics and Automation

Angadi Institute of Technology and Management, Belagavi

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DEPARTMENT OF ROBOTICS AND AUTOMATION ENGINEERING

Certificate

This is to certify that Project entitled **"MULTI- LANGUAGE INTERACTIVE TICKET DISPENSING ROBOT"** is work carried out by **Ms. Vaishnavi Patil (2AG23RA413), Ms. Bhavani Revankar (2AG23RA401), Ms. Shilpa Shettar (2AG22RA008) and Ms. Soumya Biradar (2AG22RA010),** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Robotics and Automation Engineering** under **Visvesvaraya Technological University, Belagavi** during the year **2025 - 2026**. It is certified that all the correction/suggestion indicated for internal assessment have been incorporated in the report. The Project Phase report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering Degree.

……..	……..	……..	……
Signature of the Guide	Signature of the Coordinator	Signature of the HOD	Signature of the Principal
Dr. Pooja Anagolkar	Prof. Suneel Nagavi	Dr. Pooja Anagolkar	Dr. Anand Deshpande
Professor & HOD	Assistant Professor,	Professor & HOD	Principal and Director
Dept. of R&A, AITM	Dept. of R&A, AITM	Dept. of R&A, AITM	AITM, Belagavi.

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DECLARATION

We, **Ms. Vaishnavi Patil (2AG23RA413), Ms. Bhavani Revankar (2AG23RA401), Ms. Shilpa Shettar (2AG22RA008) and Ms. Soumya Biradar (2AG22RA010)** studying in the 7th semester of Bachelor of Engineering in Robotics and Automation Engineering at Angadi Institute of Technology and Management, Belagavi, hereby declare that this project work entitled **"Multi- Language Interactive Ticket Dispensing robot"** which is being submitted by us in the partial fulfillment for the award of the degree of **Bachelor of Engineering in Robotics and Automation Engineering** from **Visvesvaraya Technological University, Belagavi** is an authentic record of us carried out during the academic year **2024-2025** under the guidance of **Dr. Pooja Angolkar,** Department of Robotics and Automation Engineering, Angadi Institute of Technology and Management, Belagavi.

We further undertake that the matter embodied in the dissertation has not been submitted previously for the award of any degree or diploma by us to any other university or institution.

Date: 15/12/2025

Place: Belagavi

Ms. Vaishnavi Patil 2AG23RA413

Ms. Bhavani Revankar 2AG23RA401

Ms. Shilpa Shettar 2AG22RA008

Ms. Soumya Biradar 2AG22RA010

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First and foremost, we wish to record our sincere gratitude to **Management of Angadi Institute of Technology and Management, Belagavi,** and to our beloved Principal **Dr. Anand Deshpande,** Angadi Institute of Technology and Management, Belagavi for his constant support and encouragement in preparation of this report and for making available library and laboratory facilities needed to prepare this report.

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Date: 15/12/2025

Place: Belagavi

Ms. Vaishnavi Patil 2AG23RA413

Ms. Bhavani Revankar 2AG23RA401

Ms. Shilpa Shettar 2AG22RA008

Ms. Soumya Biradar 2AG22RA010

ABSTRACT

This project introduces a **"Multi-Language Interactive Ticket Dispensing Robot"** designed to automate ticket booking in public transport environments using voice commands. Built around a Raspberry Pi and speech recognition engine, the system enhances user convenience and accessibility by supporting four regional languages English, Hindi, Kannada, and Marathi. The interaction begins with the detection of a predefined wake word, following which the system prompts the user to select a language. Once selected, all subsequent voice interactions occur in the chosen language, enabling users to specify the destination and number of tickets without manual input.

After confirming the travel details via voice, the system generates a QR code for payment. A dummy mobile application simulates the payment process for demonstration purposes. Once the payment is verified, a thermal printer connected to the hardware prints the travel ticket. The solution offers a low-cost, user-friendly, and inclusive approach to ticket vending that is especially beneficial in regions with linguistically diverse populations, helping to reduce queues and promote contactless transactions in public transport systems.

Keywords:

Voice-based ticketing, multilingual speech recognition, Raspberry Pi, smart transport systems, QR code payment, thermal printer, human-computer interaction, regional language support. etc

TABLE OF CONTENTS

Declaration		i
Acknowledgement		ii
Vision & Mission		iii
Abstract		iv

INDEX

CHAPTER 1	**INTRODUCTION**	**1**
	1.1 Problem definition	2
	1.2 Project motivation	3
CHAPTER 2	**LITERATURE SURVEY**	**4**
CHAPTER 3	**OBJECTIVES**	**11**
CHAPTER 4	**WORKING PRINCIPLE**	**13**
CHAPTER 5	**METHODOLOGY**	**14**
CHAPTER 6	**HARDWARE AND SOFTWARE USED**	**15**
	6.1 Hardware Requirements	19
	6.2 Software Requirements	20
CHAPTER 7	**IMPLEMENATATION**	**21**
	7.1 System Architecture Overview	22
	7.2 Module-Wise Implementation	24
	7.3 Flow of the Implementation	24
	7.4 Hardware Design	30
CHAPTER 8	**APPLICATIONS, ADVANTAGES AND DISADVANTAGES**	**31**
CHAPTER 9	**IMPLEMENTATION OF PYTHON PROGRAM**	**60**
CHAPTER 10	**RESULT AND SNAPSHOTS**	**64**
	CONCLUSION	**65**
	REFERENCES	**68**

LIST OF FIGURES

SL. NO	LIST OF FIGURES	Page No.
4.1.1	Block Diagram	12
6.1.1	Raspberry Pi (1 GB RAM or higher)	15
6.12	USB Microphone	16
6.1.3	Speaker	16
6.1.4	Thermal Printer	17
6.1.5	LCD / TFT Touch Display	17
6.1.6	Battery Pack / Power Adapter	18
6.1.7	MicroSD Card(16GB or more)	18
7.1.1	Hardware Design Sections	21
7.1.2	Voice interaction and ticket details flow diagram	25
7.1.3	A flow diagram Payment confirmation and ticket printing	26

LIST OF SNAPSHOTS

SL. No	List of Snapshots	Page No.
1.	Front View	61
2.	Display Unit	62
3.	Ticket vending	62
4.	Top View	63
5.	Back view	64

CHAPTER 1

INTRODUCTION

The automation of public transport services has become a key step in the implementation of smart cities and intelligent transportation systems, which are changing the world. To make public transport systems more efficient, accessible, and user-friendly, one of the steps is to automate public transport ticketing systems. The manual ticketing process in public transport is one of the most time-consuming and labor-intensive activities, and it is particularly burdensome in public transport facilities like bus stops, metro stations and railway stations where demand is very high. Ticket counters and machines still do not cater to people who are not familiar with the interface, have little literacy, or are more interested in contactless services after the pandemic period.

The Indian population is huge and diverse in languages and cultures. Conventional ticket vending machines and mobile applications may have the option of using English and Hindi, but there are still millions of people who prefer to communicate in their mother tongue which can be Telugu, Bengali, Gujarati, and many more. The non-existence of multilingual support gives rise to complaints and limits the usage of the service among the people living in remote areas and around the country. The project settles this issue by creating a multilingual voice ticket machine enabling customers to talk to the machine and be understood.

The system, built on Raspberry Pi, uses speech recognition and text-to-speech (TTS) technologies to create an easy user interface that can recognize and reply to commands in the following languages: English, Hindi, Kannada, and Marathi. The system aims to reduce the hassle of booking tickets for everyone, especially senior citizens, tech-illiterate individuals, and those with physical or visual disabilities.

What is presented here is going to be a machine for ticket issuance with the integrated hardware, which will be controlled via voice commands. A Raspberry Pi is the central processing unit selected here primarily due to its low cost, versatility, and wide range of supported programming languages and libraries. The system incorporates a wake word detector that sets off the user interaction as soon as someone gets near the kiosk. The moment the wake word is recognized, a microphone is turned on, and the user is asked to pick a language from the available ones.

At the language selection moment, the system will continue in that language talking to the user to get the necessary travel details like destination and the number of tickets. The input will be taken care of by speech recognition, which will then dissect the utterance to recognize the action. When the user has provided all the necessary information, a QR code is created instantly and shown, which the user can then scan to have the payment done through a mock mobile payment application created just for this demonstration.

Once the payment is confirmed, the Raspberry Pi sends a command to the connected thermal printer (either through serial or USB) to print the ticket. The whole process is intended to be smooth, so no physical contact and no manual typing are required. This voice-first interface makes the system extremely convenient in very busy public places, thus providing quicker service and less human interaction.

The testing of the system has been done with different accents and voices for the four languages supported. The system's modular design makes it easy to add other languages in the future. As a result, the payment integration could be developed further to include real UPI, NFC, or card-based transactions, which would make it applicable in actual transportation networks

1.1PROBLEM DEFINITION:

Public transport systems in India still rely heavily on manual and touchscreen-based ticketing methods, which often lead to long queues, human errors, delays, and accessibility issues, especially for the elderly, visually impaired, and users unfamiliar with English or digital interfaces. These challenges have been further intensified by the COVID-19 pandemic, which highlighted the need for contactless and safer ticketing solutions. Existing automated systems largely fail to support regional languages or provide inclusive access for all users. To address these issues, a voice-controlled ticket vending system is proposed that enables ticket booking through natural voice interaction in Indian languages such as English, Hindi, Kannada, and Marathi. By integrating speech recognition, wake-word detection, and QR code–based digital payment, the system allows passengers to complete the entire ticketing process without physical touch or human intervention, thereby improving convenience, safety, efficiency, and inclusivity in public transport services across India.

1.2PROJECT MOTIVATION:

The rationale behind creating a Multi-Language Voice-Based Ticket Vending System is the necessity of enhancing the public transport system in a multilingual country like India through its efficiency, accessibility, and user-friendliness. The existing methods of ticketing that consist of manual counters and touchscreen vending machines are not only the longest and human-error-susceptible but also overlook the linguistic diversity and digital literacy gap among the population. In several territories of India, users prefer to communicate in their local languages such as Kannada, Marathi, or Hindi rather than English. Nevertheless, most of the automated systems either provide only one language, English, or have very limited interfaces in regional languages, making them practically impossible for the majority of the commuters, particularly in rural or semi-urban zones, to use. Also, seniors, visually impaired, and those not acquainted with smartphones often run into problems while trying to access digital ticketing platforms. These obstacles point out an urgent requirement for more inclusive public transport systems.

The other aspect of the pandemic and COVID-19 was that it altered the public's expectation regarding the presence of contactless, touch-free interactions, particularly in bus and railway stations that attract large crowds.

This has further made it necessary to come up with systems that facilitate users to express themselves through speech without touching surfaces that are used by others. The use of voice recognition and voice command technology has been the solution to the problem as it enables completely hands-free ticket booking.

CHAPTER 2

LITERATURE REVIEW

[1] Title: Automated RFID-Based Ticket Vending System for Public Transportation
Authors: Mathankumar et al.
Journal: International Journal of Engineering Research and Technology (IJERT)
Summary: Mathankumar et al. proposed an automated ticket vending machine for public buses using RFID-based smart cards to replace traditional paper tickets and reduce manual intervention. The system employs a PIC microcontroller to manage core operations such as fare deduction and journey validation. IR sensors are used to detect passenger entry and exit, while load sensors help verify passenger presence, ensuring accurate fare calculation. By utilizing smart cards, the system provides a secure, contactless, and efficient ticketing solution, improving transaction speed and minimizing human errors in public transportation systems.

[2] Title: Solar-Powered Voice-Enabled Smart Vending Machine for Green Buildings
Authors: Bisht et al.
Journal: International Journal of Scientific Research in Engineering and Management (IJSREM)
Summary: Bisht et al. presented a solar-powered, voice-enabled vending machine designed for integration within green building ecosystems. The system leverages Machine Learning for accurate voice recognition and the Internet of Things (IoT) for intelligent monitoring and control. By operating without direct human interaction, the vending machine enhances hygiene and user convenience. Its solar energy utilization emphasizes sustainability and resource efficiency, while flexible IoT-based control enables seamless operation in smart environments. The proposed design highlights an eco-friendly and intelligent approach to automated vending systems.

[3] Title: Low-Cost Voice-Operated Vending Machine Using Speech Recognition
Authors: Surjit et al.
Journal: International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering (IJAREEIE)
Summary: Surjit et al. developed a low-cost voice-operated vending machine utilizing the V3 speech recognition module to interpret spoken user commands. The system converts voice inputs into control signals that trigger the dispensing of items. By using minimal hardware components and a simple control architecture, the design reduces overall system cost and complexity. The proposed solution is particularly suitable for semi-urban and rural areas, where affordability, ease of use, and accessibility are critical factors.

[4] Title: AI-Based Personal Assistant Using Speech Recognition and NLP
Authors: Venkatesh Kumar et al.
Journal: International Journal of Computer Applications (IJCA)
Summary: Venkatesh Kumar et al. developed an AI-based personal assistant system intended for computer-based applications.

The system integrates speech recognition and Natural Language Processing (NLP) to understand and respond to user commands in a conversational manner. Although designed mainly for desktop environments, the proposed architecture provides important insights into voice-driven system design, command processing, and user interaction models. These concepts can be effectively adapted to embedded and automated systems such as voice-based ticket vending machines, enhancing usability and interaction efficiency.

[5] Title: Smart Card and GSM-Based Automatic Ticket Vending System
Authors: Visveswaran et al.
Journal: International Journal of Engineering and Advanced Technology (IJEAT)
Summary: Visveswaran et al. designed an automatic ticket vending system using RFID-based smart cards and GSM communication. The system identifies passengers through RFID readers and transmits ticketing data using ZigBee technology. By minimizing human intervention, the approach improves transaction speed, accuracy, and efficiency in public transportation environments.

[6] Title: Voice Recognition-Based Automated Appointment Booking System
Authors: Namdev et al.
Journal: International Journal of Computer Science and Information Technologies (IJCSIT)
Summary: Namdev et al. introduced an automated appointment booking system that utilizes voice recognition and Automatic Speech Recognition (ASR). Designed for healthcare scheduling, the system reduces manual workload and enhances customer experience. The work demonstrates the effectiveness of speech-based automation in time-sensitive and critical service applications.

[7] Title: RFID and RTC-Based Smart Card Automatic Ticket Vending Machine
Authors: Kale et al.
Journal: International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE)
Summary: Kale et al. developed a smart card-based automatic ticket vending machine using RFID and Real-Time Clock (RTC) modules. Users select destinations and check bus availability through a keypad interface. The system improves ticketing automation and can be further enhanced by integrating voice interfaces to increase accessibility for diverse users.

[8] Title: Speech-Based Intelligent Billing System Using Multi-Model Voice Recognition
Authors: Dhinavahi et al.
Journal: International Journal of Advanced Computer Science and Applications (IJACSA)
Summary: Dhinavahi et al. implemented a speech-based billing system incorporating Google's Speech-to-Text API along with custom-trained Convolutional Neural Networks (CNNs) for dialect recognition.
The system emphasizes adaptability across different languages and accents, making it highly relevant for multilingual environments such as voice-based ticket vending systems.

[9] Title: RFID and ZigBee-Based Voice-Enabled Smart Ticket Vending System with GPS
Authors: Bhuvaneswari et al.
Journal: International Journal of Scientific and Engineering Research (IJSER)
Summary: Bhuvaneswari et al. explored a ticket vending system integrating RFID authentication, ZigBee communication, and a voice-enabled GPS module for route guidance. The system also includes alcohol detection for safety enhancement. This comprehensive approach improves automation, commuter safety, and accessibility in public transport systems.

[10] Title: NLP-Based Chatbot for Voice-Enabled Ticket Booking
Authors: Rawal et al.
Journal: International Journal of Emerging Technologies and Innovative Research (IJETIR)
Summary: Rawal et al. proposed a chatbot-based ticket booking system using Natural Language Processing (NLP) integrated with Dialogflow. Users can interact via voice or text to make bookings. The system highlights scalable, cloud-based conversational interfaces suitable for multilingual and large-scale public service applications.

[11] Title: RFID and ZigBee-Based Intelligent Ticket Vending and Bus Tracking System
Authors: Jagtap et al.
Journal: International Journal of Engineering Research and Technology (IJERT)
Summary: Jagtap et al. presented an RFID and ZigBee-based ticket vending system featuring automatic fare deduction, GSM-based real-time bus tracking, and voice-enabled GPS guidance. The system also integrates alcohol sensors to enhance safety. This solution improves commuter convenience, operational efficiency, and real-time transport management.

[12] Title: Ticket Vending Machine for the Visually Impaired Persons
Authors: Rehmat Baddar et al.
Journal: Diva Portal (Academic Thesis)
Summary: Rehmat Baddar et al. presented a ticket vending machine designed specifically for visually impaired users. The system focuses on accessibility by integrating voice guidance and tactile inputs to help users complete ticket transactions independently. Audio prompts guide users through ticket selection and payment processes, reducing dependency on human assistance. This work highlights the importance of inclusive design and serves as a foundational reference for voice-based and assistive ticket vending systems.

[13] Title: Online Chatbot Based Ticketing System for Museums Using Machine Learning
Author: Syed Beeban Basha
Journal: Computer Science Journal
Summary: This paper proposes an online chatbot-based ticketing system for museums using machine learning techniques. The chatbot interacts with users in natural language, assisting in ticket booking, availability checks, and query resolution. The system

improves user engagement, reduces staff workload, and provides a scalable digital ticketing solution. The approach is relevant for conversational and AI-driven ticket vending systems.

[14] Title: Real-Time Human–Machine Interaction Through Voice Augmentation Using Artificial Intelligence
Authors: M. N. Sumaiya et al.
Journal: Springer
Summary: The authors explored real-time voice-based human–machine interaction using artificial intelligence. The system enhances machine responsiveness by integrating speech recognition, voice augmentation, and AI decision logic. This research supports the development of interactive voice-controlled systems and strengthens the foundation for multilingual voice-based ticket vending machines.

[**15] Title:** Smart Digital Terminal Devices with Speech Recognition and Speech Control
Authors: Mirko Sajić et al.
Journal: 10th Mediterranean Conference on Embedded Computing (MECO)
Summary: This paper discusses smart digital terminal devices equipped with speech recognition and speech control capabilities. The system architecture focuses on embedded platforms and real-time processing of voice commands. The study is highly relevant for designing embedded voice-controlled kiosks and ticket vending terminals.

[16] Title: Voice Controlled Home Automation System
Authors: Ribil Mary Roy, Blessy Sabu, Nisha A, Aneesh R. P.
Journal: ICBSII
Summary: The authors developed a voice-controlled home automation system using speech recognition and embedded controllers. Voice commands are processed to control electrical appliances efficiently. Although applied to home automation, the same voice command processing and control logic can be adapted for ticket vending and kiosk automation systems.

[17] Title: A Novel Approach of Automation in Ticket Vending Machine Using ARM Processor
Authors: S. Sukhumar et al.
Journal: International Journal of Engineering Trends and Technology (IJETT)
Summary: This work presents an automated ticket vending machine using an ARM processor for efficient control and processing. The system reduces manual ticketing errors and improves transaction speed.
The research highlights the importance of embedded processors in modern ticket vending machines and supports integration with voice-based modules.

[18] Title: Audio-Enabled Smart Parking Ticket Generation System with Driver Face Identification Using AI
Authors: Jahanvi Purohit, Suresh Kumar Gawre

Journal: Springer
Summary: The authors proposed a smart parking ticket generation system integrating audio interaction and AI-based face identification. Voice prompts assist users, while facial recognition ensures secure identification. This system demonstrates how AI, voice interfaces, and automation can be combined for smart ticket generation applications.

[19] Title: Voice Interaction Recognition Design in Real-Life Scenario Mobile Robot Applications
Authors: Shih-An Li et al.
Journal: Applied Sciences (MDPI)
Summary: This study focuses on designing voice interaction recognition systems for mobile robots operating in real-life environments. It emphasizes noise handling, speech accuracy, and real-time response. The findings are useful for improving speech recognition reliability in public-place ticket vending systems.

[20] Title: Web Based Voice Assistant for Railways Using Deep Learning Approach
Authors: Prasad Vadamodula, R. Cristin, T. Daniya
Journal: International Conference on Ubiquitous Computing and Intelligent Information Systems
Summary: The paper introduces a web-based voice assistant for railway services using deep learning. Users can interact through voice commands to obtain ticket and travel information. This system demonstrates the effectiveness of voice-based interfaces in large-scale public transportation services.

[21] Title: Virtual Bank Assistance: An AI-Based Voice Bot for Better Banking
Author: Sarath Chandra Oruganti
Journal: International Journal of Research
Summary: This research presents an AI-based voice bot for banking services. The system automates customer interactions through speech recognition and AI responses. The architecture and interaction model can be adapted to ticket vending and customer assistance kiosks.

[22] Title: Web Based Voice Assistant for Railways Using Deep Learning Approach
Authors: Prasad Vadamodula et al.
Journal: Springer
Summary: This work extends voice assistant functionality using deep learning for railway ticketing services. It supports voice queries, booking assistance, and real-time responses. The system is highly relevant for multilingual voice-based ticket vending solutions.

[23] Title: Use of a Cellular Phone in Mobile Robot Voice Control
Authors: T. Kubik, M. Sugisaka
Journal: IEEE

Summary: This early research explored mobile robot control using voice commands transmitted via cellular phones. The study laid the groundwork for remote voice control systems and highlights the long-term applicability of voice interfaces in automated systems.

[24] Title: The Service Revolution: Intelligent Automation and Service Robots
Authors: Jochen Wirtz et al.
Summary: This paper discusses the impact of intelligent automation and service robots in modern service industries. It emphasizes AI-driven customer interaction and automation, supporting the adoption of smart ticket vending machines in public services.

[25] Title: The Strategic Impacts of Intelligent Automation for Knowledge and Service Work
Author: Dhaya Sindhu Battina
Summary: The author analyzed how intelligent automation transforms service work by improving efficiency and reducing operational costs. This study supports the implementation of AI-based ticket vending systems in transportation and public services.

[26] Title: Introduction to AI Chatbots
Authors: Aishwarya Gupta et al.
Summary: This paper introduces AI chatbot fundamentals, including NLP and conversational interfaces. The concepts discussed are useful for integrating chatbot and voice-assistant features into smart ticket vending systems.

[27] Title: Robotic Automation Process – The Operating System for the Digital Enterprise
Author: Sorin Anagnoste
Summary: This work explains robotic process automation and its role in digital enterprises. The principles discussed support backend automation and intelligent control in ticket vending systems.

[28] Title: Approaches to Designing the Personality of Robots and Its Effects on Human–Robot Interaction
Authors: Yi Mou et al.
Summary: The authors studied how robot personality design influences human interaction. This research is useful for designing user-friendly voice responses in smart ticket vending machines.

[29] Title: The Service Robot Revolution
Authors: Stefanie Paluch et al.
Summary: This publication explores the rapid growth of service robots in customer-facing applications. It highlights the role of AI, automation, and voice interaction, reinforcing the relevance of smart vending systems.

[30] Title: Distant Speech Recognition
Authors: Matthias Wölfel, John McDonough
Summary: This book focuses on distant speech recognition techniques, including noise suppression and microphone array processing. These concepts are essential for improving voice recognition accuracy in noisy public environments like bus stations.

CHAPTER 3

OBJECTIVES

1.Unveiling a multilingual voice-based interface for public transport ticket booking that permits users to interact naturally in the local languages, apart from English; Hindi; Kannada; and Marathi.

2.To create a system for the detection of a wake-word that starts the system only when the user is ready to interact, thus allowing efficient hands-free operation even in noisy places where the public is.

3.To ensure smooth ticket generation and payment processing through the integration of voice commands with QR code generation for digital payment and ticket printing via a thermal printer.

4.To make public transport more accessible and inclusive by introducing an easy way for elderly travelers, blind people, and those who can't read or don't know much about technology to use it.

5.To eliminate human involvement and minimize contact in the ticket booking process leading to a redirection of wait times, operational errors, and a safer, contactless travel experience.

CHAPTER 4

WORKING PRINCIPLE

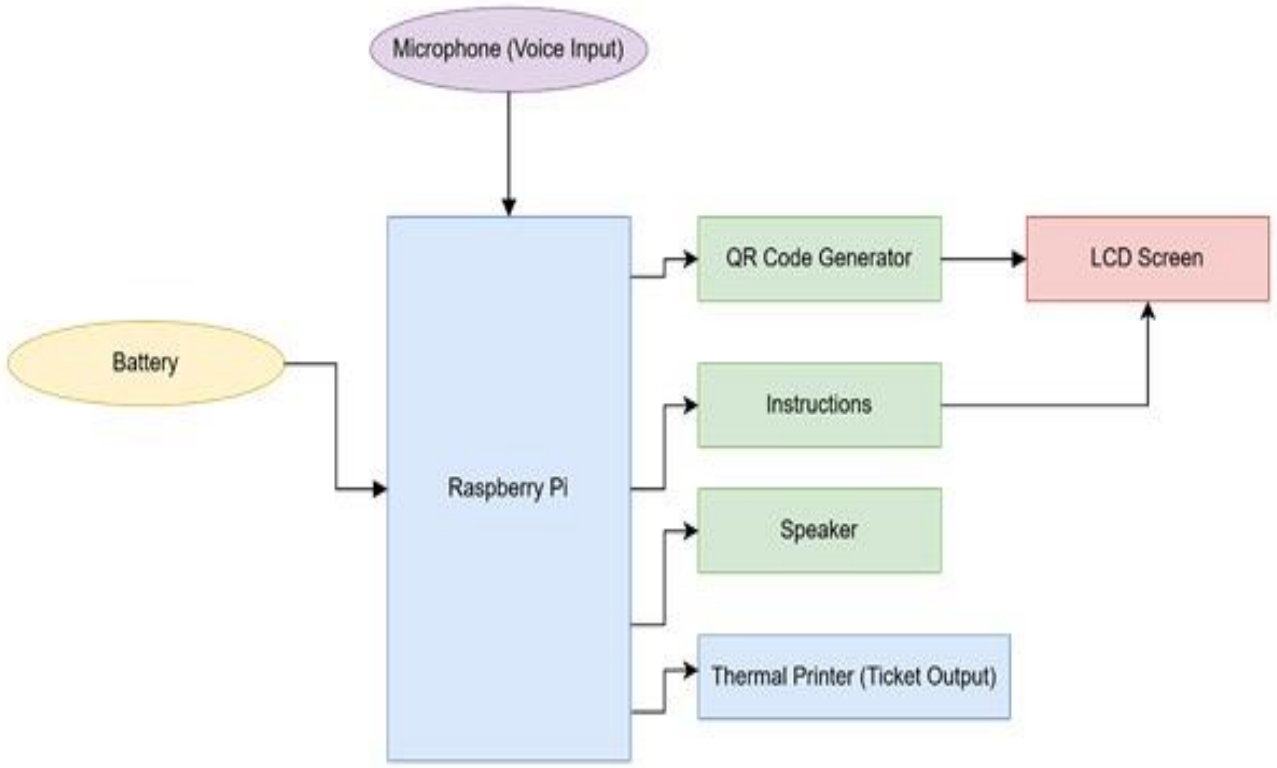


Fig 4.1 Block Diagram

The Multi-Language Voice-Based Ticket Vending System is based on speech recognition and voice-guided interaction that is realized through a Raspberry Pi microcontroller. The system listens continuously for a predetermined wake word (e.g., "Hello Ticket") via a voice recognition engine. After the wake word is recognized, the system turns on the microphone and starts a dialogue with the user.

The user has the option to choose from English, Hindi, Kannada, or Marathi as the preferred language through voice. As soon as the language is chosen, all subsequent interactions are conducted in the selected language with the help of text-to-speech (TTS) and speech-to-text (STT) technologies.

The system will then inquire about the user's destination and the number of tickets they want. The speech recognition module will process these inputs and the user will be asked to confirm them to prevent mistakes.

After confirmation, the system will create a QR code for the ticket payment which will then be displayed on a screen.

A dummy payment app (for demo purposes) simulates the transaction. On successful payment verification, the system generates a travel ticket through an attached thermal printer.

The whole thing is completely touch-free, voice-controlled, and language-adaptive, which makes it accessible to a large number of users, even those who cannot read or are not familiar with digital interfaces. The system unites hardware (Raspberry Pi, mic, thermal printer) and software (speech processing, QR code creation, payment simulation) to provide a fast and inclusive public ticketing service.

CHAPTER 5

METHODOLOGY:

The multi-language voice-based ticket vending system has been developed using a systematic and modular approach. The objective of such an approach is to provide a powerful machine, a system that is adaptable to languages and a hardware-software combo that has no difficulties in integration. The complete procedure can be further broken down into the following stages:

5.1 The design of the system and setup of the hardware: The main hardware consists of Raspberry Pi, which is the brain behind the entire system. The peripheral devices, namely the USB microphone, speaker, thermal printer, and LCD display, are all connected to the Raspberry Pi. The user's voice input is captured by the microphone, and the speaker plays back the sound. The thermal printer is connected to the system for ticket printing, and the LCD is utilized for showing visual prompts and instructions regarding the payment.

5.2 Wake word detection and selection of the language: The system stays silent until the specified wake word is recognized, using either a very simple voice detection or a wake-word recognition module. After being turned on, the system asks the user to choose a language. The language selected is recorded in audio and is then processed through the Speech Recognition library, in collaboration with Google Speech-to-Text APIs, which allows for multilingual recognition and hence the processing of the audio in different languages.

5.3 Voice-Based Ticket Booking Interaction: Once the language has been selected, the system continues to communicate with the user in the chosen language. The user is asked to vocalize the destination name and the number of tickets they want. These answers are recorded and transcribed from voice to text. Also, accuracy checks and confirmation loops are used to confirm and check the user input before moving to the next step.

5.4 QR Code Generation and Payment Handling: A ticket price is determined according to the user's answers, and a matching QR code is created with the help of either the PyQRCode or qrcode library. This QR code appears on the LCD screen for the UPI-enabled demo payment application to scan. The system then creates and delivers the ticket after confirming the dummy payment (which is simulated in this project).

5.5 Ticket Printing and Session Termination: The Raspberry Pi issues a print command to the thermal printer upon receipt of a payment confirmation and a ticket is printed with details of the destination, the number of tickets, the amount paid, and the time of issuing. The system then returns to its starting state and awaits the next user interaction after doing the printing.

CHAPTER 6

HARDWARE AND SOFTWARE USED:

An integrated combination of hardware components and software tools is the foundation of the Multi-Language Voice-Based Ticket Vending System development, which successfully incorporates voice interaction, multilingual processing, simulation of payment, and ticket printing. The description of hardware modules and software frameworks crucial to the system will be given in this chapter, and their roles in making the ticket vending operations seamless and real-time will be explained.

6.1 Hardware Requirements

The hardware components that were chosen for the system make sure that the system works reliably, performs efficiently, and is compatible with the speech-based human-machine interaction. The modules contribute individually to the overall system of the kiosk and also enable the application to run in real time.

6.1.1 Raspberry Pi (1 GB RAM or higher)

Acts as a CPU and runs the Python application, speech recognition and driving the GUI are executed, and peripheries are controlled by it. The GPIO pins provide interfacing at the hardware level with the buzzer and push buttons.



Fig 6.1.1 Raspberry Pi (1 GB RAM or higher)

6.1.2 USB Microphone

Grabs voice messages from the user such as wake words, language selection, the destination, and the number of tickets. To ensure accurate speech recognition in many languages, high-quality audio capture is a must.



Fig 6.1.2 USB Microphone

6.1.3 Speaker

IT Text-to-Speech responses that provide auditory feedback in the preferred language. This feature was developed for people with poor literacy skill or visual impairments in it.



Fig 6.1.3 Speaker

6.1.4 A Thermal Printer

Prints the ticket after payment confirmation using ESC/POS commands. It is of the highest reliability and will dispense the ticket outputs to an extremely great kiosk level at a significant speed.



Fig 6.1.4 A Thermal Printer

6.1.5 LCD / TFT Touch Display

A 3.5 inch screen compatible with Raspberry Pi was used to exhibit interactive GUI elements like welcome messages, payment QR codes, and ticket details as well as confirmation screens.



Fig 6.1.5 LCD / TFT Touch Display

6.1.6 Battery Pack / Power Adapter

Raspberry Pi and other components get constant 5V power from this. A 9V external power supply is utilized



Fig 6.1.6 Battery Pack / Power Adapter

6.1.7 MicroSD Card (16 GB or more)

It saves the Raspberry Pi operating system, Python software, libraries, and system settings. Sufficient memory space guarantees the OS operation and logging without interruptions.



Fig 6.1.7 MicroSD Card (16 GB or more)

6.1.8 Push Buttons (Optional)

Utilized for hand resets, interactions with maintenance, or input overrides as an option.

6.1.9 Connecting Wires and Breadboard / PCB

Make it possible to organize and even do electrical connections safely to the various components such as GPIO wiring, printer connections, buzzer circuits, and expansion headers. A custom PCB design also aids in reliability and cable management for Kiosk installations, which is another advantage Point.

6.2 Software Requirements

The software stack was created to handle real-time speech interaction, GUI display, QR code generation, payment communication, and hardware control. The following tools and libraries are the main components of the system's primary software ecosystem.

6.2.1 Raspberry Pi OS

The operating system that runs the whole application. It provides support for Python, device drivers, networking, and USB peripherals.

6.2.2 Python 3.x

The main programming language that was used to create the whole ticket vending application including GUI, speech processing, payment communication, and printing.

6.2.3 SpeechRecognition Library

It does the speech-to-text conversion for the wake words and user commands. It supports multilingual recognition by using Google Speech-to-Text APIs.

6.2.4 gTTS (Google Text-to-Speech)

It creates spoken replies in English, Hindi, Kannada, and Marathi making sure that the system is able to interact with people from different language backgrounds.

6.2.5 PyQRCode or qrcode Library

It creates QR codes in real-time based on the destination, ticket amount, and transaction details allowing the payment simulation process.

6.2.6 Pillow (PIL)

It is used to handle images for GUI display, especially QR codes and interface graphics.

6.2.7 RPi.GPIO or gpiozero

It gives access to Raspberry Pi’s GPIO pins for buzzer, push button, and optional relay output control.

6.2.8 Pygame

It is the software that allows for the audio playback of generated TTS files, facilitating smooth and non-blocking sound output within the Python application.

6.2.9 Internet Connection

It is needed for speech recognition (Google STT), translation, and MQTT cloud communication. It guarantees real-time voice processing and payment confirmation.

CHAPTER 7

IMPLEMENTATION:

The primary focus of the Multi-Language Voice-Based Ticket Vending System is to combine several technologies including speech interaction, multilingual processing, graphical user interface management, IoT-based payment confirmation, and thermal ticket printing, all under a Raspberry Pi setting. In this chapter, the total system implementation is reported including architectural choices, module-wise development, communication flows, and execution logic. The system is designed as a real-time, fully automated ticket vending kiosk that can interact with the users only by speech while at the same time providing a very responsive visual interface on a 3.5-inch display of Raspberry Pi.

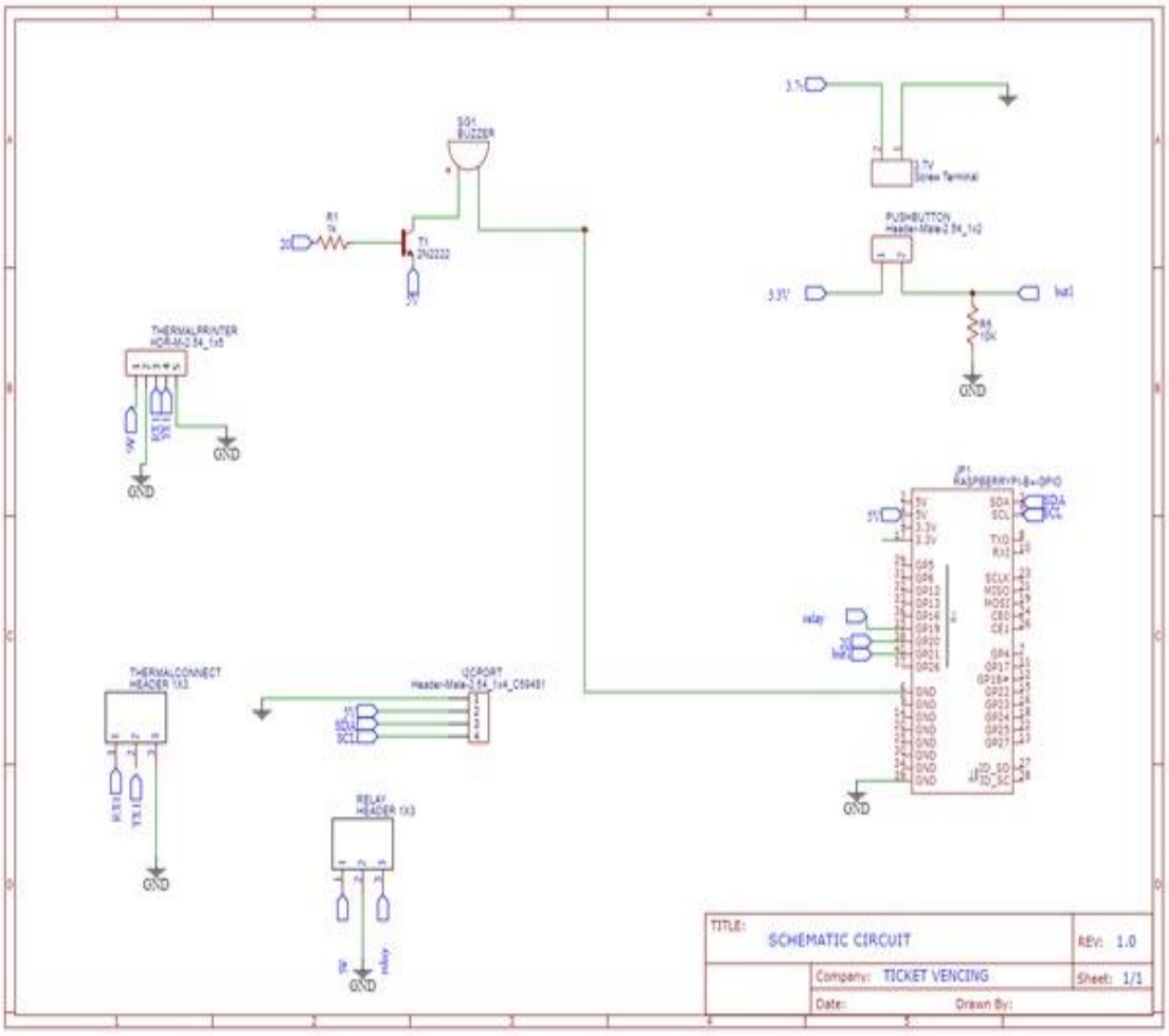


Fig 7.1 hardware design sections

7.1 System Architecture Overview

The system architecture as shown in fig 7.1 consists of four layers that are solidly united, thus ensuring modularity, maintenance simplicity, and the possibility of enlarging the system in the future.

(a) User Interaction Layer

It is this layer that takes charge of the total visual and auditory communication with the user. The 3.5-inch Raspberry Pi display runs a Tkinter-based multi-screen GUI which helps the user step-by-step through the process of ticket vending. The graphical display provides the user with a series of visual options such as home, language selection, ticket entry, QR code display, and payment confirmation. A Text-to-Speech (TTS) operation that makes use of gTTS works together with the visual interface by giving verbal instructions in the chosen language and thus, making the service-user friendly people that are not proficient in reading or visually impaired.

(b) Speech Processing Layer

This layer is in charge of carrying out all the operations related to voice. The Raspberry Pi is provided with the Speech Recognition library for the purpose of recording and interpreting user speech. A user can get the system to take action only after the wake word that has been set previously (“hello ticket”) is recognized. This system, which listens passively, permits activation by touchless means. The system, then, to understand the user inputs such as language preference, destination, and ticket count, resorts to multilingual speech recognition. Google STT provides strong recognition support for all of the above languages, which include English, Hindi, Kannada, and Marathi.

(c) Payment Processing and IoT Communication Layer

The layer enables the payment validation process which is secured and performed in real-time through MQTT in a way that is similar to the application of a digital workflow in the UPI or wallet transaction involving the customer and the merchant at the purchase point in the physical store, i.e., after the QR code is generated, the system will be waiting for a payment confirmation message which will be published by the dummy mobile app on a specific MQTT topic. The Raspberry Pi is going to be listening to that topic and will be changing the payment status based on the receipt of a "success" or "failure" message. Only after receiving a "success" message will the ticket be printed by the system.

(d) Ticket Generation and Thermal Printing Layer

The last layer takes care of the ticket printing process by the RP205 USB thermal printer and it does so by issuing commands based on the ESC/POS protocol. The communication between the printer and the Raspberry Pi is made possible by the python-escpos library. Once the payment is done, the system prints out a ticket containing the required details such as the destination, the number of tickets, the time of issue, and a thank-you note. The printer is able to cut the paper automatically, so it corresponds to the quality of a commercial ticket kiosk.

7.2 Module-Wise Implementation

The system is built on the principles of modular programming, which means that every functional block can work alone and at the same time keep up with the interaction of the other blocks.

7.2.1 Graphical User Interface using Tkinter

The graphical user interface is designed with Tkinter frames that represent different screens:

• Home Screen

• Language Selection Screen

• Ticket Confirmation Screen

• QR Code Payment Screen

• Payment Status Screen

A custom screen controller is loading and unloading frames dynamically to ensure smooth transitions. The GUI is designed according to the 480×320 resolution of the 3.5-inch display.

7.2.2 Wake Word Activation

The system is always on and ready to respond to the wake word "hello ticket." The Speech Recognition module is in polling mode, using the live microphone input. As soon as the phrase is recognized, the system moves from the home screen to the language selection process. This interaction model eliminates the use of physical buttons and encourages hygienic, touch-free operation in public places.

7.2.3 Multilingual Speech Recognition and Language Handling

When users give a command, they can say "language" to indicate their desired language. Subsequently, the system waits for a voice command like "English," "Hindi," "Kannada," or "Marathi." The system detects the language and corresponds the spoken one to its ISO code for languages, thus switching the TTS output to the chosen one. In instances where there is no clear speech, English will be the default language. This multilingual interaction framework is a good way to provide inclusivity among various user groups.

7.2.4 Ticket Input Capture (Destination and Quantity)

After determining the language, the system asks for the destination. The recognized speech is then transcribed to text in the selected language. Following that, the user indicates how many tickets are needed. The system validates and shows the inputs on the GUI. Subsequently, the system double-checks the captured information via on-screen text and TTS output.

7.2.5 QR Code Generation for Digital Payment

The system creates a unique QR code that integrates the destination, ticket quantity, and a timestamp. The QR code appears prominently on the Tkinter screen for easy scanning with the mock payment app. QR code generation is executed through the qrcode library and is perfectly customized for the screen size to guarantee clarity.

7.2.6 MQTT Payment Confirmation

The Raspberry Pi, through HiveMQ broker, subscribes to a specific MQTT topic (like "ticket/payment"). The dummy application sends out JSON messages such as:

{"payment": "success"}

or

{"payment": "failed"}

The system is always listening to this topic. It will only move to the printing phase after receiving "success". Otherwise, a failure message is shown and heard, and the system is turned back into standby mode.

7.2.7 Thermal Ticket Printing (RP205 USB)

The RP205 thermal printer uses ESC/POS commands to perform the printing process. With the help of the python-escpos library, it allows:

• Text formatting

• Line spacing adjustments

• Automatic cutting of paper the ticket printed consists of the following:

• System header

• Destination

• Number of tickets purchased

• Transaction timestamp

• Thank-you message

Once done, the system goes back to the home screen and waits for the next user action.

7.3 Flow of the Implementation:

The complete implementation flow is as follows:

1. The system remains on stand by till the wake word is spoken.

2. The user speaks “hello ticket.”

3. The user then says “language” to change the language.

4. The system takes note of the language selection.

5. The system inquires the destination.

6. The system proceeds to ask the number of tickets.

7. A QR code is generated for payment by the system.

8. The user makes the payment through the dummy app.

9. The payment is confirmed by MQTT.

10. The ticket is printed out using the RP205 printer.

11. The system is reset and ready for the next customer.

The illustration below depicts the flow chart of the implementation:

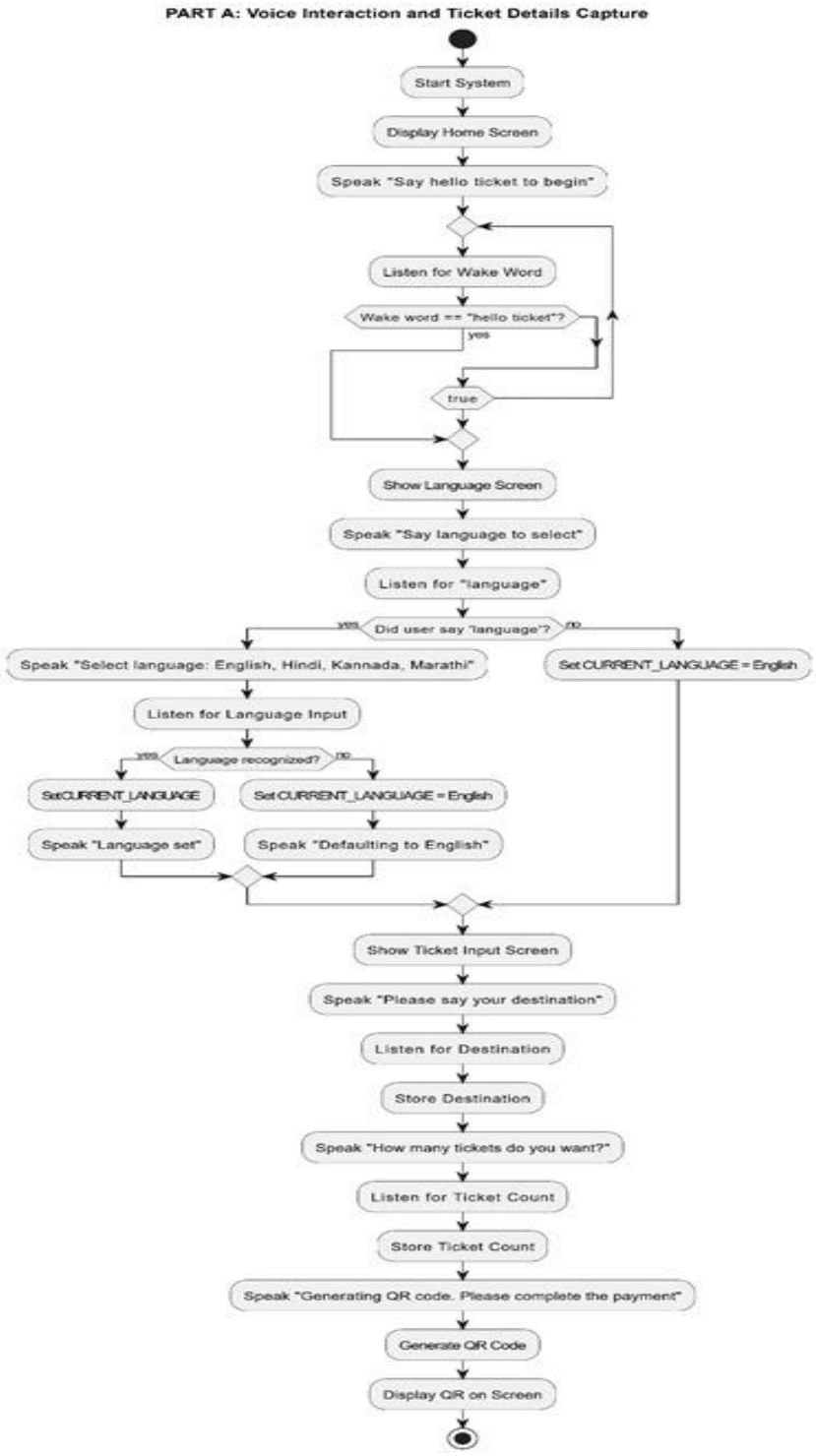


Fig 7.3(a) voice interaction and ticket details flow diagram

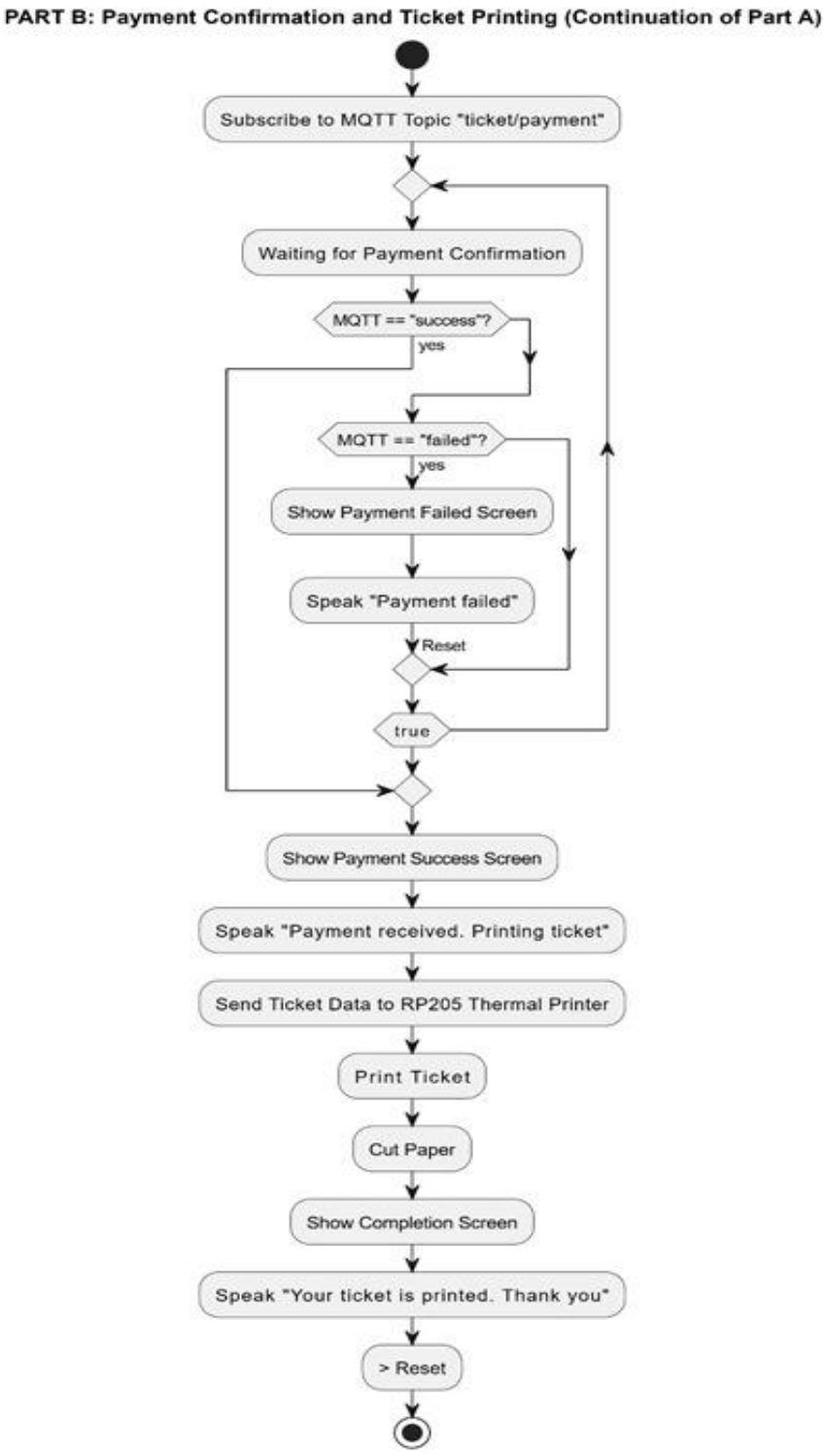


Fig 7.3(b) continuation of part A flow diagram

Payment confirmation and ticket printing

7.4 Hardware Design:

The hardware architecture of the Multi-Language Voice-Based Ticket Vending System integrates the Raspberry Pi 3, along with a thermal printer for tickets, a buzzer for alerts, user input through push buttons, and other connecting ingredients like I²C communication headers and relay output terminals. The design guarantees that power distribution is safe, signal isolation is achieved, and PCB routing is done cleanly so that the kiosk operates reliably. This part of the paper provides an overview of each subsystem alongside the schematic's translation into a functional PCB.

7.4.1. Core Processing Unit: Raspberry Pi GPIO Interface

The Raspberry Pi is the main controller of the whole vending machine. The GPIO header (J1) in the schematic provides:

• 5V and 3.3V supply lines

• Ground reference

• GPIO pins for the control of the buzzer, for pushbutton input, and for relay activation

• I²C lines (SDA, SCL) for the future expansion of sensors or displays

The PCB layout interconnects these pins to specified connectors and components while observing the required trace separation and reducing the noise on the power lines to a minimum.

The Raspberry Pi has the following key duties:

• Running the Tkinter graphical user interface

• Handling the speech recognition and TTS

• Managing the payment confirmation through MQTT

• Issuing commands in ESC/POS to the thermal printer

• Controlling the buzzer and taking pushbutton input

7.4.2. Thermal Printer Interface Circuit

Two connectors are present:

a. THERMALPRINTER Header (Power + UART signals)

The thermal printer has the following requirements:

• a high-current 9V external power supply

• RX/TX pins for communication with the Raspberry Pi (if UART mode is utilized)

• GND reference

In your schematic:

• The 9V input is routed separately to achieve cleaner power delivery.

• The ground is shared between the Pi and the printer to assure signal integrity.

• The UART lines TX/RX are optional depending on the choice of USB or TTL mode.

b. THERMALCONNECT Header(power +UART signals)

This header divides the printer connector into neat, identifiable pins.

It facilitates wiring during the testing period and provides the option of replacing the printer module without the need for soldering changes.

On the PCB:

• The thermal connector is located near the edge for easy access to ribbon connectors.

• 9V power traces are made wider to accommodate high current spikes during printing.

7.4.3. Buzzer Driver Circuit (Transistor-Based Switching)

An SG1 buzzer is controlled by a transistor (2N2222) in a low-side switching configuration. Description of the Circuit Operation •GPIO → R1 (1 kΩ) → Base of T1 (2N2222)

• Collector connected to buzzer negative terminal

• Buzzer positive connected to 5V supply When GPIO is HIGH:

• 2N2222 is in saturation

• Buzzer is fed with complete 5V

• Audible sound is produced When GPIO is LOW:

• Transistor is in OFF state

• Buzzer is not producing sound On PCB:

• Buzzer is mounted on the top edge for sound projection and the PCB has thick traces for stable current flow

• The transistor is close to the buzzer so that there is no inductive noise

7.4.4. Push Button Input Circuit

Pushbutton (PUSHBUTTON) is connected in a pull-down manner:

• One terminal to 3.3V

• Other terminal to GPIO with a series resistor

• 10kΩ resistance (R5) when the button is not pressed, pulls the signal to GND Working Logic

• GPIO reads LOW when the button is released

• GPIO reads HIGH when the button is pressed This guarantees a debounced digital input for:
• Reset signal

• Manual override

• Future maintenance functions On PCB:

• The button is located near the top corner for effortless access during kiosk mounting panel installation

7.4.5. I²C Expansion Header

The I2CPORT header has the following signals:

• SDA

• SCL

• 5V

• GND

Thus, it permits to add more hardware in the future like:

• OLED display

• RTC module

• NFC/QR scanner

• More sensors

For the external wiring the PCB places this connector at the left edge.

7.4.6. Power Distribution

The PCB contains:

• 9V input terminal for the thermal printer

• 5V rail from Raspberry Pi for logic circuits

• Clear power separation to prevent noise from Pi GPIO

The PCB is designed with thick traces for:

• 9V high-current paths

• Ground return paths

• Thermal printer power lines

Thus, operation is guaranteed to be stable even at heavy load printing cycles.

7.4.7. PCB Layout Design Considerations

The layout is improved to:

a. Minimal Crosstalk

Printer power lines, logic-level signals, and audio/buzzer traces are routed separately.

b. Component Placement

• Raspberry Pi GPIO header is centrally placed for shortest routing.

• Printer connector is placed near output slot.

• Buzzer is on the top side for clear audio.

• Pushbutton is positioned for user access.

c. Ground Plane

A full bottom-layer ground pour enhances signal integrity, EMI reduction, and heat dissipation.

d. Trace Optimization

• Power traces are wide.

• Signal traces are short.

• No sharp 90° turns are done.

• Via count is minimized.

The PCB screenshot clearly demonstrates proper routing discipline and clean separation of power and logic sections. The hardware design sections is as shown below:

CHAPTER 8

APPLICATIONS, ADVANTAGES AND DISADVANTAGES:

Applications:

1. Bus stands, railway stations, metro stations for ticket dispensing.
2. Integrated into smart kiosks for multilingual and contactless ticketing.
3. Useful in areas with diverse linguistic tourists.
4. Helps non-tech-savvy and illiterate users access digital ticketing systems.
5. For domestic travelers preferring voice over touchscreen interfaces.
6. Can be used to issue entry tokens for seminars or events.
7. Ticket/token generation for shuttle services.

Advantages:

1. Accommodates users in English, Hindi, Kannada, Marathi.
2. Reduces contact, ensuring hygiene (post-pandemic use).
3. No need to read or type; voice-only interaction.
4. Minimizes the need for manned counters.
5. Voice input is quicker than navigating menus.
6. Raspberry Pi-based solution is affordable.
7. Can be deployed at multiple locations with low maintenance.

Disadvantages:

1. Requires internet for Google Speech APIs.
2. May misinterpret commands in noisy environments.

CHAPTER 9

IMPLEMENTATION OF PYTHON PROGRAM

```
import time
import datetime
import random
import wave
import json
import os
import socket

import datetime, time, json
from googletrans import Translator
from gtts import gTTS
import pygame
import pyaudio
import speech_recognition as sr
import qrcode
import subprocess
from escpos.printer import Usb
import paho.mqtt.client as mqtt
import tkinter as tk
from PIL import Image, ImageTk

time.sleep(10)

languages = ["en-IN", "mr-IN", "hi-IN", "kn-IN"]
defaultlanguage = "en-IN"
currentlanguage = "english"
MIC_DEVICE_INDEX = 3
```

```
AUDIO_SECONDS = 4
AUDIO_FILE = "ticket_in.wav"

translator = Translator()
recognizer = sr.Recognizer()

pygame.mixer.init()

ORIGIN_CITY = "Belagavi"

DEST_FARES = {
    "kolhapur": 150,
    "nipani": 60,
    "dharwad": 120,
    "hubli": 140,
    "kittur": 80,
    "sankeshwar": 90,
    "pune": 550,
    "bengaluru": 700,
    "mysore": 650,
    "mangalore": 750,
    "mumbai": 900,
    "delhi": 1800,
    "belagavi": 0
}


DEST_DISPLAY = {
    "kolhapur": "Kolhapur",
    "nipani": "Nipani",
```

```
    "dharwad": "Dharwad",
    "hubli": "Hubli",
    "kittur": "Kittur",
    "sankeshwar": "Sankeshwar",
    "pune": "Pune",
    "bengaluru": "Bengaluru",
    "mysore": "Mysuru",
    "mangalore": "Mangaluru",
    "mumbai": "Mumbai",
    "delhi": "Delhi",
    "belagavi": "Belagavi"
}

def get_ip_address():
    try:
        s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
        s.connect(("8.8.8.8", 80))
        ip = s.getsockname()[0]
        s.close()
        return ip
    except:
        return "IP not found"

def get_all_destinations_text():
    try:



 return ", ".join(DEST_DISPLAY.values())
    except:
        return ""
```

```
def get_printer():
    p = Usb(0x0fe6, 0x811e, in_ep=0x82, out_ep=0x01)
    p._raw(b"\x1b\x40")
    return p

mqtt_client = mqtt.Client()
current_waiting_ticket_id = None
paid_ticket_id = None

def on_mqtt_message(client, userdata, msg):
    global paid_ticket_id
    try:
        payload = msg.payload.decode().strip()
        print("MQTT RECV:", payload)
        if payload.startswith("PAID:"):
            tid = payload.split(":", 1)[1].strip()
            print("MQTT Payment confirm received for:", tid)
            paid_ticket_id = tid
    except Exception as e:
        print("Error in MQTT message:", e)

def init_mqtt():
    mqtt_client.username_pw_set(MQTT_USER, MQTT_PASS)
    mqtt_client.on_message = on_mqtt_message



    mqtt_client.connect(MQTT_BROKER, MQTT_PORT, 60)
    mqtt_client.subscribe(MQTT_SUB_TOPIC)
    mqtt_client.loop_start()
```

```
try:
    root = tk.Tk()
    root.title("Hello Ticket")
    root.geometry("480x320")
    root.configure(bg="white")
    root.attributes("-fullscreen", True)
    root.bind("<Escape>", lambda e: root.attributes("-fullscreen", False))

    lbl_status = tk.Label(
        root,
        text="Starting...",
        font=("Arial", 40, "bold"),
        fg="black",
        bg="white",
        wraplength=440,
        justify="center"
    )
    lbl_status.pack(pady=5)

    lbl_info = tk.Label(
        root,
        text="",
        font=("Arial", 30, "bold"),
        fg="black",


 bg="white",
        wraplength=440,
        justify="center"
    )
```

```
lbl_info.pack(pady=5)

qr_label = tk.Label(root, bg="white")
qr_label.pack(pady=5)

def ui_update(status=None, info=None):
    if status is not None:
        lbl_status.config(text=status)
    if info is not None:
        lbl_info.config(text=info)
    try:
        root.update_idletasks()
        root.update()
    except:
        pass

def ui_show_qr(path):
    if not os.path.exists(path):
        return
    try:
        img = Image.open(path)
        img = img.resize((300, 300))
        img_tk = ImageTk.PhotoImage(img)
        qr_label.config(image=img_tk)


qr_label.image = img_tk
        ui_update()
    except Exception as e:
        print("UI QR error:", e)
```

```
  def ui_clear_qr():
        qr_label.config(image="", text="")
        qr_label.image = None
        ui_update()

except Exception as e:
    print("UI init error:", e)
    root = None
    def ui_update(status=None, info=None): pass
    def ui_show_qr(path): pass
    def ui_clear_qr(): pass

def speak(text):
    global currentlanguage
    if currentlanguage == "marathi":
        lang_code = "mr"
    elif currentlanguage == "hindi":
        lang_code = "hi"
    elif currentlanguage == "kannada":
        lang_code = "kn"
    else:
        lang_code = "en"
 ui_update(status=f"{currentlanguage.upper()} SPEAK", info=str(text))

    try:
        tts = gTTS(text=str(text), lang=lang_code)
    except:
        tts = gTTS(text=str(text), lang="en")

    filename = "tts_tmp.mp3"
```

```
    tts.save(filename)
    pygame.mixer.music.load(filename)
    pygame.mixer.music.play()
    while pygame.mixer.music.get_busy():
        try:
            if root:
                root.update_idletasks()
                root.update()
        except:
            pass
        time.sleep(0.05)
    try:
        pygame.mixer.music.stop()
    except:
        pass
    try:
        os.remove(filename)
    except:
        pass



def checkcurrentlanguage():
    global defaultlanguage, currentlanguage
    if defaultlanguage == "en-IN":
        currentlanguage = "english"
    elif defaultlanguage == "mr-IN":
        currentlanguage = "marathi"
    elif defaultlanguage == "hi-IN":
        currentlanguage = "hindi"
```

```
    elif defaultlanguage == "kn-IN":
        currentlanguage = "kannada"
    else:
        currentlanguage = "english"


def set_language_from_word(lang_word):
    global defaultlanguage, currentlanguage
    w = (lang_word or "").strip().lower()
    if w == "marathi":
        defaultlanguage = "mr-IN"
        currentlanguage = "marathi"
        speak("भाषा मराठीवर सेट केली आहे.")
    elif w == "hindi":
        defaultlanguage = "hi-IN"
        currentlanguage = "hindi"
        speak("भाषा हिंदी पर सेट की गई है.")
    elif w == "kannada":
        defaultlanguage = "kn-IN"
        currentlanguage = "kannada"
        speak("ಭಾಷೆಯನ್ನು ಕನ್ನಡಕ್ಕೆ ಸೆಟ್ ಮಾಡಲಾಗಿದೆ.")



  elif w == "english":
        defaultlanguage = "en-IN"
        currentlanguage = "english"
        speak("Language is set to English.")
    else:
        speak("Sorry, invalid language. Please say English, Hindi, Marathi or Kannada.")
        return False
    return True
```

```
def ask_language_selection():
    speak("Please say your language. English, Hindi, Marathi or Kannada.")
    ui_update(status="Language", info="Say: English / Hindi / Marathi / Kannada")
    txt = listen_english_once()
    if not txt:
        speak("I did not hear the language. I will keep English.")
        return
    speak(txt)
    set_language_from_word(txt)

def recordaudio(seconds=AUDIO_SECONDS, filename=AUDIO_FILE):
    form_1 = pyaudio.paInt16
    chans = 1
    samp_rate = 44100
    chunk = 4096

    audio = pyaudio.PyAudio()
    try:
        stream = audio.open(


  format=form_1,
            rate=samp_rate,
            channels=chans,
            input_device_index=MIC_DEVICE_INDEX,
            input=True,
            frames_per_buffer=chunk
        )
    except:
        return
```

```
frames = []
    start_time = time.time()

    last_ui_update = 0

    try:
        for _ in range(0, int((samp_rate / chunk) * seconds)):
            if time.time() - start_time > 5:
                break

            data = stream.read(chunk, exception_on_overflow=False)
            frames.append(data)

            elapsed = int(time.time() - start_time)
            if elapsed != last_ui_update:
                last_ui_update = elapsed
                ui_update(
                    status="Listening...",



             info=f"{elapsed} sec"
                )

    except:
        pass

    stream.stop_stream()
    stream.close()
    audio.terminate()
```

```
    ui_update(status="", info="")

    if not frames:
        return

    wf = wave.open(filename, 'wb')
    wf.setnchannels(chans)
    wf.setsampwidth(audio.get_sample_size(form_1))
    wf.setframerate(samp_rate)
    wf.writeframes(b"".join(frames))
    wf.close()

def recognize_file(language_code, filename=AUDIO_FILE):
    global recognizer
    if not os.path.exists(filename):
        return None
    with sr.AudioFile(filename) as source:
        recognizer.adjust_for_ambient_noise(source)



    audio_data = recognizer.record(source)
    try:
        return recognizer.recognize_google(audio_data, language=language_code)
    except:
        return None

def listen_in_current_language():
    global defaultlanguage, currentlanguage
    recordaudio()
    raw = recognize_file(defaultlanguage)
```

```
    if not raw:
        return None, None

    if currentlanguage == "marathi":
        en = translate_to_english(raw, "mr")
    elif currentlanguage == "hindi":
        en = translate_to_english(raw, "hi")
    elif currentlanguage == "kannada":
        en = translate_to_english(raw, "kn")
    else:
        en = raw

    return raw, en

def listen_english_once():
    recordaudio()
    txt = recognize_file("en-IN")
    return txt


def translate_to_english(text, src_lang):
    try:
        res = translator.translate(text, src=src_lang, dest="en")
        return res.text
    except:
        return text

def listen_for_wake_word():
    ui_update(status="Waiting", info="Say: Hello Ticket")
    txt = listen_english_once()
```

```
    if not txt:
        return False
    t = txt.lower()
    if "hello ticket" in t or "hallo ticket" in t or "hello tikit" in t:
        global defaultlanguage, currentlanguage
        defaultlanguage = "en-IN"
        currentlanguage = "english"
        speak("Ticket system activated.")
        return True
    return False

def parse_ticket_count(english_text):
    if not english_text:
        return None
    t = english_text.lower().strip()

    replacements = {
        "for ": "four ",



        " for": " four",
        "tree": "three",
        "free": "three",
        "too": "two",
        "to ": "two ",
    }
    for bad, good in replacements.items():
        t = t.replace(bad, good)

    import re
```

```
    m = re.search(r"\d+", t)
    if m:
        try:
            n = int(m.group(0))
            if n > 0:
                return n
        except:
            pass

    words_map = {
        "one": 1, "two": 2, "three": 3, "four": 4,
        "five": 5, "six": 6, "seven": 7, "eight": 8,
        "nine": 9, "ten": 10,
        "eleven": 11, "twelve": 12, "thirteen": 13,
        "fourteen": 14, "fifteen": 15, "sixteen": 16,
        "seventeen": 17, "eighteen": 18, "nineteen": 19,
        "twenty": 20
    }


    tokens = t.replace("-", " ").split()
    for w in tokens:
        if w in words_map:
            return words_map[w]

    for w, n in words_map.items():
        if w in t:
            return n

    return None
```

```
def detect_destination(english_text):
    if not english_text:
        return None
    t = english_text.lower()

    for key in DEST_FARES.keys():
        if key in t:
            return key
    for key in DEST_FARES.keys():
        if DEST_DISPLAY[key].lower() in t:
            return key
    return None

def generate_ticket_id():
    return "TX" + str(random.randint(100000, 999999))

def generate_qr(content, filename="ticket_qr.png"):


   img = qrcode.make(content)
    img.save(filename)
    ui_show_qr(filename)

def wait_for_payment(ticket_id, timeout_sec=60):
    global paid_ticket_id

    if currentlanguage == "marathi":
        speak("कृपया मोबाईल किंवा वेब ॲपवर पे बटण दाबा.")
    elif currentlanguage == "hindi":
        speak("कृपया मोबाइल या वेब ऐप पर पे बटन दबाइए.")
```

```
    elif currentlanguage == "kannada":
        speak("ದಯವಿಟ್ಟು ಮೊಬೈಲ್ ಅಥವಾ ವೆಬ್ ಆ್ಯಪ್‌ನಲ್ಲಿ ಪೇ ಬಟನ್ ಒತ್ತಿ.")
    else:
        speak("Please press the pay button on the mobile or web app.")

    start = time.time()
    while time.time() - start < timeout_sec:
        ui_update(
            status="Waiting Payment",
            info=f"Ticket {ticket_id} | Time left: {int(timeout_sec - (time.time() - start))} s"
        )
        if paid_ticket_id == ticket_id:
            return True
        time.sleep(0.5)
    return False

def print_ticket(origin, dest_key, tickets, amount, ticket_id):


   dest_name = DEST_DISPLAY.get(dest_key, dest_key.title())
    p = get_printer()

    p.set(align="center", width=2, height=2)
    p.text("SMART BUS\n")

    p.set(align="center", width=1, height=1, text_type="B")
    p.text("VOICE TICKET SYSTEM\n")
    p.text("--------------------------------\n")

    p.set(align="left", text_type="B")
```

```
p.text(f"From     : {origin}\n")
p.text(f"To       : {dest_name}\n")
p.text("Bus No   : KA 25 AB 1234\n")
p.text("Seat No  : Auto\n\n")

p.set(text_type="NORMAL")
p.text(f"Tickets  : {tickets}\n")
p.text(f"Fare     : Rs {amount // tickets}\n")
p.text(f"Total    : Rs {amount}\n")
p.text(f"TicketID : {ticket_id}\n")

now = datetime.datetime.now().strftime("%d-%m-%Y  %H:%M:%S")
p.text(f"Date/Time: {now}\n")
p.text("--------------------------------\n")

p.set(align="center", text_type="B")
p.text("Scan Ticket\n")


qr_data = {
    "ticket_id": ticket_id,
    "from": origin,
    "to": dest_name,
    "tickets": tickets,
    "amount": amount,
}

p.qr(json.dumps(qr_data), size=6)
time.sleep(0.3)
```

```
    p.text("\nHave a Safe Journey\n")
    p.text("Thank You!\n\n\n")

    p.cut(mode="PART")

def run_ticket_flow():
    global current_waiting_ticket_id, paid_ticket_id
    global defaultlanguage, currentlanguage

    checkcurrentlanguage()

    ask_language_selection()
    checkcurrentlanguage()

    dest_key = None
    max_attempts = 3



    for attempt in range(max_attempts):
        if attempt == 0:
            if currentlanguage == "marathi":
                prompt = "कृपया तुमचे गंतव्यस्थान सांगा. कोल्हापूर, निपाणी, धारवाड आणि हुबळी हे पर्याय आहेत आणि इतर पर्याय स्क्रीनवर दिसतात."
            elif currentlanguage == "hindi":
                prompt = "कृपया अपना गंतव्य स्थान बताइए। विकल्प हैं कोल्हापुर, निपानी, धारवाड़ और हुबली, और अन्य विकल्प स्क्रीन पर दिखाई देंगे।"
            elif currentlanguage == "kannada":
                prompt = "ದಯವಿಟ್ಟು ನಿಮ್ಮ ಗಮ್ಯಸ್ಥಾನವನ್ನು ಹೇಳಿ. ಪರದೆಯ ಮೇಲೆ ಕೊಲ್ಹಾಪುರ, ನಿಪ್ಪಾಣಿ, ಧಾರವಾಡ ಮತ್ತು ಹುಬ್ಬಳ್ಳಿ ಮತ್ತು ಇತರ ಆಯ್ಕೆಗಳು ತೋರಿಸುತ್ತವೆ."
```

```
        else:
            prompt = "Please say your destination. Options are Kolhapur, Nipani, Dharwad and Hubli and others shown on screen."
    else:
        if currentlanguage == "marathi":
            prompt = "गंतव्यस्थान स्पष्ट झाले नाही. कृपया पुन्हा सांगा, किंवा कॅन्सल म्हणा, किंवा रिपीट म्हणा."
        elif currentlanguage == "hindi":
            prompt = "गंतव्य स्थान स्पष्ट नहीं हुआ. कृपया फिर से बोलिए, या कैंसल बोलिए, या रिपीट बोलिए."
        elif currentlanguage == "kannada":
            prompt = "ಗಮ್ಯಸ್ಥಾನ ಸ್ಪಷ್ಟವಾಗಲಿಲ್ಲ. ದಯವಿಟ್ಟು ಮತ್ತೆ ಹೇಳಿ, ಅಥವಾ ಕ್ಯಾನ್ಸಲ್ ಅಥವಾ ರಿಪೀಟ್ ಎಂದು ಹೇಳಿ."
        else:
            prompt = "I did not get the destination. Please say again, or say repeat to hear options, or say cancel."

    speak(prompt)

  ui_update(status="Destination",info="Say city / repeat / cancel\n\n" + get_all_destinations_text())

    dest_raw, dest_en = listen_in_current_language()
    if not dest_en:
        continue

    en_low = dest_en.lower()

    if "cancel" in en_low:
        speak("Booking cancelled. Returning to main menu.")
        ui_update(status="Cancelled", info="Say Hello Ticket to start again")
```

```
        defaultlanguage = "en-IN"
        currentlanguage = "english"
        ui_clear_qr()
        return

    if "repeat" in en_low or "again" in en_low:
        continue

    dest_key = detect_destination(dest_en)
    if dest_key:
        break

if not dest_key:
    speak("Sorry, I could not understand the destination. Returning to main menu.")
    ui_update(status="Error", info="Destination not understood")
    defaultlanguage = "en-IN"
    currentlanguage = "english"


  ui_clear_qr()
    return

fare_one = DEST_FARES[dest_key]
dest_name = DEST_DISPLAY[dest_key]

tickets = None
max_attempts = 3

for attempt in range(max_attempts):
    if attempt == 0:
```

```
        if currentlanguage == "marathi":
            prompt = "किती तिकीटे हवी आहेत ते सांगा."
        elif currentlanguage == "hindi":
            prompt = "कितने टिकट चाहिए, कृपया बताइए."
        elif currentlanguage == "kannada":
            prompt = "ಎಷ್ಟು ಟಿಕೆಟ್ ಬೇಕು ಎಂದು ಹೇಳಿ."
        else:
            prompt = "Please say the number of tickets."
    else:
        if currentlanguage == "marathi":
            prompt = "टिकिटांची संख्या समजली नाही. कृपया पुन्हा सांगा, किंवा कॅन्सल म्हणा."
        elif currentlanguage == "hindi":
            prompt = "टिकट की संख्या समझ में नहीं आई. कृपया फिर से बोलिए या कैंसल बोलिए."
        elif currentlanguage == "kannada":
            prompt = "ಟಿಕೆಟ್ ಸಂಖ್ಯೆ ಸ್ಪಷ್ಟವಾಗಿಲ್ಲ. ದಯವಿಟ್ಟು ಮತ್ತೆ ಹೇಳಿ ಅಥವಾ ಕ್ಯಾನ್ಸಲ್ ಎಂದು ಹೇಳಿ."
        else:


    prompt = "I did not understand the ticket count. Please say again, or say cancel."


    speak(prompt)
    ui_update(status="Tickets", info="Say number / cancel")


    t_raw, t_en = listen_in_current_language()
    if not t_en:
        continue
    en_low = t_en.lower()


    if "cancel" in en_low:
```

```
        speak("Booking cancelled. Returning to main menu.")
        ui_update(status="Cancelled", info="Say Hello Ticket to start again")
        defaultlanguage = "en-IN"
        currentlanguage = "english"
        ui_clear_qr()
        return

    tickets = parse_ticket_count(t_en)
    if tickets and tickets > 0:
        break

if not tickets:
    speak("Sorry, I could not understand the ticket count. Returning to main menu.")
    ui_update(status="Error", info="Ticket count not understood")
    defaultlanguage = "en-IN"
    currentlanguage = "english"
    ui_clear_qr()


 return

amount = tickets * fare_one
ticket_id = generate_ticket_id()

max_attempts = 3
confirmed = False

for attempt in range(max_attempts):

    if attempt == 0:
```

```
        if currentlanguage == "marathi":
            summary = f"आपण {ORIGIN_CITY} ते {dest_name} साठी {tickets} तिकीटे निवडली आहेत. एकूण रक्कम {amount} रुपये आहे. कृपया पुष्टी करा."
        elif currentlanguage == "hindi":
            summary = f"आपने {ORIGIN_CITY} से {dest_name} के लिए {tickets} टिकट चुने हैं. कुल राशि {amount} रुपये है. कृपया पुष्टि कीजिए."
        elif currentlanguage == "kannada":
            summary = f"ನೀವು {ORIGIN_CITY} ರಿಂದ {dest_name} ಗಮ್ಯಸ್ಥಾನಕ್ಕೆ {tickets} ಟಿಕೆಟ್ ಆಯ್ಕೆ ಮಾಡಿದ್ದೀರಿ. ಒಟ್ಟು ಮೊತ್ತ {amount} ರೂಪಾಯಿ. ದಯವಿಟ್ಟು ದೃಢೀಕರಿಸಿ."
        else:
            summary = f"You selected {tickets} ticket or tickets from {ORIGIN_CITY} to {dest_name}. Total amount is rupees {amount}. Please confirm."
    else:
        if currentlanguage == "marathi":
            summary = "कृपया हो किंवा नाही असे सांगा. किंवा कॅन्सल म्हणा."
        elif currentlanguage == "hindi":
            summary = "कृपया हाँ या नहीं बोलिए. या कैंसल बोलिए."
        elif currentlanguage == "kannada":


            summary = "ದಯವಿಟ್ಟು ಹೌದು ಅಥವಾ ಇಲ್ಲ ಎಂದು ಹೇಳಿ. ಅಥವಾ ಕ್ಯಾನ್ಸಲ್ ಎಂದು ಹೇಳಿ."
        else:
            summary = "Please say yes or no. Or say cancel."


    speak(summary)
    ui_update(status="Confirm", info="Say yes / no / cancel")


    raw_c, en_c = listen_in_current_language()


    if not en_c:
```

```
            continue

        en_low = en_c.lower()

        if "cancel" in en_low:
            speak("Booking cancelled. Returning to main menu.")
            ui_update(status="Cancelled", info="User cancelled")
            defaultlanguage = "en-IN"
            currentlanguage = "english"
            ui_clear_qr()
            return

        if any(x in en_low for x in ["yes", "ok", "confirm", "correct"]):
            confirmed = True
            break

        if any(x in en_low for x in ["no", "not", "wrong"]):
            speak("Booking cancelled.")


    ui_update(status="Cancelled", info="User said no")
            defaultlanguage = "en-IN"
            currentlanguage = "english"
            ui_clear_qr()
            return

     if not confirmed:
        speak("No valid confirmation. Returning to main menu.")
        ui_update(status="Cancelled", info="Confirmation failed")
        defaultlanguage = "en-IN"
```

```
        currentlanguage = "english"
        ui_clear_qr()
        return

    qr_data = {
        "ticket_id": ticket_id,
        "from": ORIGIN_CITY,
        "to": dest_name,
        "tickets": tickets,
        "amount": amount,
    }

    generate_qr(json.dumps(qr_data))

    try:
        mqtt_client.publish(MQTT_PUB_TOPIC, json.dumps(qr_data))
    except:
        pass



    if currentlanguage == "marathi":
        speak("कृपया QR कोड स्कॅन करून तिकीट ॲप किंवा वेबसाईटवर उघडा.")
    elif currentlanguage == "hindi":
        speak("कृपया QR कोड स्कैन करके टिकट ऐप या वेबसाइट पर खोलिए.")
    elif currentlanguage == "kannada":
        speak("ದಯವಿಟ್ಟು QR ಕೋಡ್ ಸ್ಕ್ಯಾನ್ ಮಾಡಿ ಟಿಕೆಟ್ ಆ್ಯಪ್ ಅಥವಾ ವೆಬ್‌ಸೈಟ್ ತೆರೆಯಿರಿ.")
    else:
        speak("Please scan the QR code to open the ticket in the app or website.")
```

```
    paid_ticket_id = None
    current_waiting_ticket_id = ticket_id

    if not wait_for_payment(ticket_id, timeout_sec=20):

        if currentlanguage == "marathi":
            speak("ठरलेल्या वेळेत पेमेंट झाले नाही. व्यवहार रद्द केला आहे.")
        elif currentlanguage == "hindi":
            speak("निर्धारित समय में पेमेंट नहीं हुआ. ट्रांज़ैक्शन रद्द कर दिया गया है.")
        elif currentlanguage == "kannada":
            speak("ನಿಗದಿತ ಸಮಯದಲ್ಲಿ ಪಾವತಿ ಆಗಲಿಲ್ಲ. ವ್ಯವಹಾರ ರದ್ದುಗೊಳಿಸಲಾಗಿದೆ.")
        else:
            speak("Payment not received in time. Transaction cancelled.")

        ui_update(status="Payment Timeout", info="Returning to main menu")
        ui_clear_qr()
        defaultlanguage = "en-IN"
        currentlanguage = "english"


    return

    if currentlanguage == "marathi":
        speak("पेमेंट झाले. तिकीट प्रिंट होत आहे.")
    elif currentlanguage == "hindi":
        speak("पेमेंट हो गया. टिकट प्रिंट हो रहा है.")
    elif currentlanguage == "kannada":
        speak("ಪಾವತಿ ಯಶಸ್ವಿಯಾಗಿದೆ. ಟಿಕೆಟ್ ಮುದ್ರಿಸಲಾಗುತ್ತಿದೆ.")
    else:
        speak("Payment confirmed. Printing your ticket.")
```

```
 ui_update(status="Printing", info=f"Ticket ID: {ticket_id}")
    print_ticket(ORIGIN_CITY, dest_key, tickets, amount, ticket_id)

    if currentlanguage == "marathi":
        speak("धन्यवाद. मी पुढच्या प्रवाशासाठी तयार आहे.")
    elif currentlanguage == "hindi":
        speak("धन्यवाद. मैं अगले यात्री के लिए तैयार हूँ.")
    elif currentlanguage == "kannada":
        speak("ಧನ್ಯವಾದಗಳು. ಮುಂದಿನ ಪ್ರಯಾಣಿಕರಿಗಾಗಿ ನಾನು ಸಿದ್ಧನಿದ್ದೇನೆ.")
    else:
        speak("Thank you. I am ready for the next passenger.")

    ui_clear_qr()
    ui_update(status="Ready", info="Say Hello Ticket for next passenger")

    defaultlanguage = "en-IN"
    currentlanguage = "english"


def main():
    init_mqtt()
    checkcurrentlanguage()

    global defaultlanguage, currentlanguage
    defaultlanguage = "en-IN"
    currentlanguage = "english"
    ip = get_ip_address()
    ui_update(
        status="Device IP Address",
        info=f"{ip}\nConnect via VNC"
```

```
    )
    time.sleep(6)
    ui_update(status="Loading...", info="")

    speak("Hello, this is a smart multilingual voice ticket system. Say hello ticket to start.")
    ui_update(status="Base Screen", info="Say: Hello Ticket")

    try:
        while True:
            defaultlanguage = "en-IN"
            currentlanguage = "english"
            if listen_for_wake_word():
                run_ticket_flow()
            else:
                time.sleep(0.5)
    except KeyboardInterrupt:
        pass


    finally:
        try:
            mqtt_client.loop_stop()
        except:
            pass

if __name__ == "__main__":
    main()
```

CHAPTER 10

RESULTS AND SNAPSHOTS



Fig 10.1 Front View

1. Front View

- The **front view** shows:
  - Display unit at the top
  - Ticket printer/output slot in the center
  - Clean and compact rectangular design
- Designed for **user-facing interaction**.
- Front panel is smooth to avoid accidental damage.
- Suitable for public places such as:
  - Bus stations
  - Railway stations
  - Colleges
  - Museums



Fig 10.2 Display Unit

2. Display Unit (Touch Display)

- The display unit is mounted at the **top front section** of the robot.
- It consists of a **TFT/LCD screen** protected by a transparent cover.
- Used to show:
  - Language selection menu
  - Ticket options (source, destination, number of tickets)
  - Fare details
  - Payment status
  - System messages and instructions
- The display enables **human–machine interaction (HMI)** and supports **touch or button-based input**.
- Designed for easy visibility at standing height.



Fig 10.3 ticket vending

3. Ticket Vending Unit (Printer Section)

- The **ticket vending module** is placed in the **middle front section**.
- A **thermal ticket printer** is used to generate tickets quickly and silently.
- Features:
  - Automatic ticket printing after successful payment
  - Paper roll housed safely inside the enclosure
  - Ticket output slot for easy collection
- Supports printed details such as:

- Date & time
- Ticket ID
- Fare amount
- Route or service information

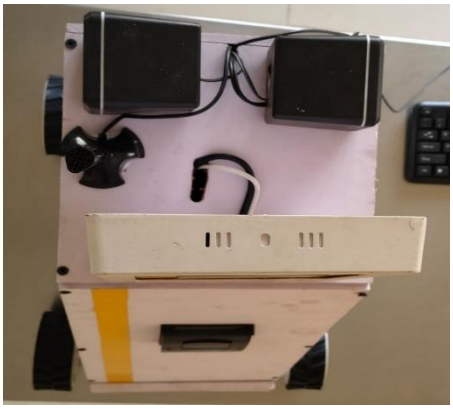


Fig 10.4 Top View

4. Top / Point View (Upper Section)

- The top section houses:
  - Speakers (left and right)
  - Microphone module
  - Communication components (Wi-Fi / Bluetooth antenna)
- Enables:
  - Clear voice interaction
  - Noise-reduced audio output
- Positioning ensures:
  - Better sound projection
  - Accurate voice recognition



Fig 10.5 Back view

5. Back View

- The **back view** contains:
  - Speaker units for **voice guidance**
  - Microphone for **voice commands**
  - Control panel access door
  - Power supply and wiring connections
- Speakers provide:
  - Multi-language voice instructions
  - Error alerts and confirmations
- Rear panel allows **easy maintenance and troubleshooting**.

CHAPTER 11

CONCLUSION AND FUTURE SCOPE

The Multi-Language Voice-Based Ticket Vending System is a brilliant example of a modern, inclusive, and automated system that uses technology altogether in public transport ticketing. This is all made possible through the incorporation of speech recognition, multi-language interaction, QR-based digital payments, and thermal ticket printing in a small kiosk powered by Raspberry Pi. The voice-activated interface aids in erasing the barriers met by the illiterate, elderly, and technology-unfamiliar passengers, thereby providing equitable access to people speaking different languages. The automation covers the complete process, thereby substantially eliminating human workload at ticket drawers, speeding up service, and providing a clean, no-touch alternative that is ideal for areas with a lot of people. The system is still very much compatible with the existing smart city infrastructure and future public transport services as it clearly showcases a powerful, extensive, and practical solution for such a scenario.

Integrating real-time digital payment platforms such as UPI, NFC, or card-based transactions, and thus replacing the demo payment mechanism currently used is one of the future enhancements of the ticket vending machine system. The speech engine update can go as far as incorporating offline, neural-network-based wake-word detection, and multilingual speech-to-text models so as to guarantee uninterrupted operation even in low-connectivity environments. Moreover, the system can also be enhanced with additional features like, for instance, facial recognition for user authentication, dynamic fare calculation based on GPS or route databases, cloud-based ticket analytics, and IoT monitoring for device health, which can all greatly contribute to increased system intelligence and usability. Thanks to the scalability of hardware and modular software design, the system can turn into a multifunctional smart kiosk that is compatible with public services applications including buses, metro stations, and railways.

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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1. Apply the fundamentals of mathematics, science and mechanical engineering concepts to analyze, formulate, and solve complex engineering problems in industries, research and academia.

PEO2. Demonstrate the ability to design and develop efficient, reliable and innovative mechanical systems that are environmentally sustainable, socially responsible and aligned with industry standards.

PEO3. Employ modern engineering tools, techniques and emerging technologies in collaborating multidisciplinary domains to address industrial and societal challenges.

PEO4. Exhibit ethical values, effective communication, resilience and leadership qualities while pursuing a professional career contributing to Sustainable Development Goals (SDGs) and social well-being.

PEO5. Understand the efficacy of lifelong learning, and develop managerial and entrepreneurial abilities to become competent professionals and industry leaders at a global level.

COURSE OUTCOMES

At the end of the Course, the Student will be able to:

CO1 : Understand the relevance and complexity of recent trends in the particular field of Engineering through literature survey.

CO2 : Able to communicate effectively.

CO3 : Understand the concept, application and demonstrate the Project implementation. **CO4 :** Acquire the management skills like budgeting and finance, planning, team works and leadership qualities.

PROGRAM SPECIFIC OUTCOMES (PSO's)

PSO 1: Use basic maths, science, electronics, programming, and introductory robotics knowledge to understand and design simple robotic and automation systems.

PSO 2: Apply advanced skills in robotics, automation, IoT, and mechatronics to create practical and innovative solutions for industry, research, and startups.

PSO 3: Develop good teamwork, communication, leadership, ethics, and management skills to work responsibly for the benefit of society.

PROGRAM OUTCOMES [PO]

PO1. Engineering Knowledge: Apply the Knowledge of Mathematics, Science, Engineering Fundamentals, and an Engineering specialization to the solution of complex Engineering problems.

PO2. Problem Analysis: Identify, Formulate, Review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, natural sciences and engineering sciences.

PO3. Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental conditions.

PO4. Conduct investigations on complex problems: Use research based knowledge and research methods including design of Experiments, analysis and interpretation of data, and synthesis of Information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate technique, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess society, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

CO-PO MAPPING

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	**2**	**3**	**2**	**2**	**1**	**–**	**–**	**–**	**–**	**1**	**–**	**1**
CO2	**2**	**2**	**3**	**2**	**2**	**–**	**1**	**–**	**–**	**1**	**2**	**–**
CO3	**2**	**2**	**2**	**3**	**3**	**–**	**1**	**–**	**–**	**1**	**1**	**–**
CO4	**1**	**–**	**–**	**–**	**–**	**2**	**2**	**3**	**3**	**3**	**3**	**2**

CO-PSO MAPPING

COs / PSOs	PSO1	PSO2	PSO3
CO1	3	3	2
CO2	2	2	2
CO3	3	2	3
CO4	2	3	3

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Signature of the guide
Dr. Pooja Angolkar
Professor and HOD
Dept. of R&A, AITM

…….

Signature of the guide
Dr. Pooja Angolkar
Professor and HOD
Dept. of R&A, AITM